# 🚀 One-Click Deployment Guide

Deploy your Lead Enrichment Web App to a live, publicly accessible website in minutes!

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## ⚡ Quick Deploy Options

Choose one platform below and click the deploy button. All platforms offer **free tiers** perfect for team use.

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## 🟢 Option 1: Railway (Recommended - Easiest)

**Why Railway?**
- ✅ Fastest deployment (2 minutes)
- ✅ Free $5/month credit (enough for this app)
- ✅ Automatic HTTPS
- ✅ Built-in database
- ✅ Modern dashboard

### Deploy Steps:

1. **Push to GitHub** (if not already):
   ```
   bash
   cd lead_enrichment_app
   git init
   git add .
   git commit -m "Lead Enrichment App"
   gh repo create lead-enrichment-app --public --source=. --push
   ```

2. **One-Click Deploy**:

   Deploy on Railway

   - Click the button above
   - Sign in with GitHub
   - Select “Deploy from GitHub repo”
   - Choose your `lead-enrichment-app` repository
   - Click “Deploy”
   - Wait 2 minutes ⏱️
   - Get your live URL! 🎉

1. **Your app will be live at**: `https://your-app-name.up.railway.app`

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# 🔵 Option 2: Render (Great Free Tier)

**Why Render?**
- ✅ Generous free tier
- ✅ Automatic SSL
- ✅ Easy to use
- ✅ Good performance

## Deploy Steps:

1. **Push to GitHub** (if needed):
   ```
   bash
   cd lead_enrichment_app
   git init
   git add .
   git commit -m "Lead Enrichment App"
   gh repo create lead-enrichment-app --public --source=. --push
   ```

2. **One-Click Deploy**:

   Deploy to Render

   - Click the button above
   - Sign in with GitHub
   - Click “New +” → “Web Service”
   - Connect your GitHub repository
   - Render will auto-detect settings from `render.yaml`
   - Click “Create Web Service”
   - Wait 3-5 minutes ⏱️
   - Get your live URL! 🎉

1. **Your app will be live at**: `https://your-app-name.onrender.com`

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# 🟣 Option 3: Heroku (Classic & Reliable)

**Why Heroku?**
- ✅ Industry standard
- ✅ Reliable uptime
- ✅ Easy scaling
- ⚠️ Requires credit card (free tier available)

## Deploy Steps:

1. **Install Heroku CLI**:
   ```
   bash
   curl https://cli-assets.heroku.com/install.sh | sh
   ```

2. **Deploy**:
   ```bash
   cd lead_enrichment_app

# Login to Heroku
heroku login

# Create app
heroku create lead-enrichment-webverse

# Initialize git (if not already)
git init
git add .
git commit -m “Lead Enrichment App”

# Deploy
git push heroku main

# Open your app
heroku open
```

1. **Your app will be live at**: `https://lead-enrichment-webverse.herokuapp.com`

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## 🟠 Option 4: PythonAnywhere (Python-Specific)

**Why PythonAnywhere?**
- ✅ Python-focused platform
- ✅ Simple setup
- ✅ Free tier available
- ✅ Good for small teams

### Deploy Steps:

1. Go to PythonAnywhere
2. Sign up for a free account
3. Upload your `lead_enrichment_app` folder
4. Configure web app:
   - Python version: 3.11
   - Framework: Flask
   - Working directory: `/home/yourusername/lead_enrichment_app`
   - WSGI file: Point to `app.py`
5. Click “Reload” and your app is live!

**Your app will be live at**: `https://yourusername.pythonanywhere.com`

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## 📋 Manual Deployment (Any Platform)

If you want to deploy to another platform, use these files:

### Required Files (Already Created):

- ✅ `requirements.txt` - Python dependencies
- ✅ `Procfile` - Process configuration
- ✅ `render.yaml` - Render config
- ✅ `railway.json` - Railway config
- ✅ `app.py` - Main application (updated for PORT)

### Steps for Any Platform:

1. **Upload/Push your code** to the platform
2. **Set Python version**: 3.11
3. **Install command**: `pip install -r requirements.txt`
4. **Start command**: `gunicorn app:app`
5. **Port**: Auto-detected from environment
6. Deploy! 🚀

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## 🔧 Post-Deployment Setup

### 1. Test Your App

Visit your live URL and verify:
- ✅ Dashboard loads
- ✅ Can add new leads
- ✅ Lead enrichment works
- ✅ Emails generate correctly

### 2. Share with Your Team

Send them the live URL:
- `https://your-app-name.railway.app` or
- `https://your-app-name.onrender.com` or
- Your custom domain

### 3. Add Custom Domain (Optional)

Most platforms allow custom domains:
- Railway: Settings → Domains → Add Custom Domain
- Render: Settings → Custom Domain
- Heroku: Settings → Domains → Add Domain

Example: `leads.webverseinnovators.com`

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## 🗄️ Database Setup

The app uses SQLite by default, which works fine for small teams. For production:

### Option A: Keep SQLite (Simple)

- ✅ Works for up to 100 leads
- ✅ No extra setup needed

- ⚠️ Data resets on app restart (some platforms)

### Option B: Upgrade to PostgreSQL (Recommended for Production)

Most platforms offer free PostgreSQL:

**Railway**:
1. Click "+ New" → "Database" → "PostgreSQL"
2. Get connection URL
3. Update `app.py` to use PostgreSQL

**Render**:
1. Create new PostgreSQL database
2. Copy connection URL
3. Update `app.py` to use PostgreSQL

**Heroku**:

```
heroku addons:create heroku-postgresql:mini
```

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## 🔐 Security for Production

Before going live with real data:

### 1. Add Authentication

```
pip install flask-login
```

Add login system to protect lead data.

### 2. Environment Variables

Set these as environment variables (not in code):
- `SECRET_KEY` - For session security
- `DATABASE_URL` - If using PostgreSQL
- `ADMIN_PASSWORD` - For admin access

### 3. Enable HTTPS

All recommended platforms provide HTTPS by default ✅

### 4. Add Rate Limiting

```
pip install flask-limiter
```

Prevent abuse of your enrichment endpoint.

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# 📊 Monitoring & Maintenance

## Check App Health

- Railway: Built-in metrics dashboard
- Render: Logs & metrics in dashboard
- Heroku: `heroku logs --tail`

## Regular Backups

Export your leads database weekly:
1. Download `database/leads.db`
2. Store securely
3. Can restore anytime

## Update the App

```
# Make changes locally
git add .
git commit -m "Update feature"
git push

# App auto-deploys! 🎉
```

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# 🆘 Troubleshooting

## App won't start?

Check logs:
- Railway: Click on deployment → Logs
- Render: Logs tab
- Heroku: `heroku logs --tail`

Common issues:
- Missing `gunicorn` in requirements.txt ✅ Already added
- Wrong Python version → Set to 3.11
- Port not configured ✅ Already fixed in app.py

## Database errors?

- SQLite file missing → Auto-created on first run
- Permission errors → Check platform file permissions

## Slow performance?

- Upgrade to paid tier
- Add PostgreSQL database
- Enable caching

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# 💰 Cost Estimates

## Free Tier Limits (All Platforms):

**Railway**:
- $5/month free credit
- ~500 hours/month
- Perfect for this app ✅

**Render**:
- 750 hours/month free
- App sleeps after 15 min inactivity
- Wakes on first request (3-5 seconds)

**Heroku**:
- 1000 hours/month free (with credit card)
- Similar sleep behavior

**PythonAnywhere**:
- Always-on on paid plans
- Free tier has daily quotas

## Paid Tiers (If Needed):

- Railway: $5-10/month for always-on
- Render: $7/month for always-on
- Heroku: $7/month for always-on

**For a small team, free tier is perfect!** ✅

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# 🎯 Recommended Quick Start

**Best path for fastest deployment**:

1. ✅ **Railway** (If you have GitHub) → 2 minutes
2. ✅ **Render** (Alternative) → 5 minutes
3. ✅ **PythonAnywhere** (No git needed) → 10 minutes

Choose Railway for the fastest experience!

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# 📞 Next Steps After Deployment

1. ✅ Click your deploy button above
2. ✅ Wait for deployment to complete
3. ✅ Visit your live URL
4. ✅ Add your first lead
5. ✅ Share URL with team
6. ✅ Start enriching leads!

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## 🎉 You’re Live!

Once deployed, your team can access the app from anywhere:
- From office computers
- From home
- From mobile devices
- From anywhere with internet!

No installation needed - just share the URL! 🚀

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**Need help?** Check the logs on your platform’s dashboard or refer to their documentation.